

Real Estate SaaS Platform

Frontend Architecture, Routing, & Feature Specification (Weeks 1-3)

Project Goal: To deliver a super-fast, professional, SEO-optimized, and smoothly animated real estate marketplace prototype. This document serves as the master blueprint for client verification prior to backend integration.

1. Core Technology & UX Standards

To achieve enterprise-grade speed and Redfin/Realtor-level smoothness, the platform utilizes the following stack:

Next.js (App Router)

Tailwind CSS

Shadcn UI

Framer Motion

Sonner (Toasts)

React-Map-GL (Mapbox)

SWR / React Query

- **Super-fast Transitions:** Framer Motion drives page layout shifts (e.g., map-to-list views).
- **Professional Feedback:** Sonner handles all promise-based loading and success/error toasts smoothly.
- **SEO Supremacy:** Dynamic metadata generation and Open Graph tags for every property URL.

2. Platform Routes & Dashboards

The application is segmented into distinct Route Groups to ensure secure, scalable, and isolated workspaces.

1. Public Marketplace (**public**)

- **/** - Home/Landing Page (Hero search, featured listings, value propositions).
- **/properties** - Map-Split Search Directory (Interactive Mapbox right, list left).
- **/properties/[slug]** - Dynamic Property Detail Page (High-res gallery, specs, agent form).
- **/agents** - Directory of platform agents and brokers.
- **/agents/[id]** - Public profile for agents (reviews, active listings).
- **/pricing** - SaaS Freemium/Premium tier comparison cards.
- **/auth/login** & **/auth/register** - Unified authentication screens.

2. Client Panel (Buyer/Tenant) (**client**)

- **/dashboard/client** - Overview (Saved searches, recently viewed).
- **/dashboard/client/favorites** - Bookmarked property assets.
- **/dashboard/client/inquiries** - Messaging portal tracking offers and tour requests.
- **/dashboard/client/settings** - Profile and notification preferences.

3. Agent/Middleman Dashboard (**agent**)

- **/dashboard/agent** - Analytics Hub (Total views, CTR, active leads).
- **/dashboard/agent/properties** - Data table of own listings.
- **/dashboard/agent/properties/create** - Multi-step spec-driven creation wizard.
- **/dashboard/agent/leads** - Kanban board for incoming client inquiries.
- **/dashboard/agent/billing** - Subscription & tier limit management.

4. Super Admin / Admin Dashboard (admin)

- **/admin** - Global System Monitor (Revenue, daily active users).
- **/admin/users** - Role assignment, verification badge issuance, account bans.
- **/admin/properties** - Moderation queue (Approve, reject, feature).
- **/admin/plans** - SaaS tier editor (Limits, pricing, Stripe sync).
- **/admin/content** - CMS for Blog, FAQ, and Policies.

3. Comprehensive Property Schema (Mock Data Spec)

Sourced from industry leaders (Redfin, Realtor.com, ViaviLab), the data structure captures every critical detail required for a high-end real estate listing.

- **Base Information:** Title, Description, Transaction Type (Buy/Rent/Share), Build Year, Agent Commission Percent, Pricing & Currency.
- **Location & Mapping:** Street Address, City, State, Zip, Coordinates (Latitude/Longitude for Google Maps/Mapbox), Neighborhood.
- **Core Specifications:** Square Footage (Total & Living area), Total Rooms, Bedrooms, Bathrooms, Balconies/Patios, Floor Number.
- **Interior & Facilities:** Detailed room-by-room breakdown, Heating/Cooling systems, Flooring type, Appliance inclusion.
- **Utilities & Connectivity:** Internet availability/speed, Water, Gas, Electricity limits, Trash collection.
- **Parking:** Garage spaces, Carport, Open parking slots, Street parking rules.
- **Risk & Climate Factors:** Flood zone rating, Fire risk index, Severe weather readiness (Crucial for modern buyers).
- **Media & Presentation:** High-res Image Array, Video Tour URL, 3D/Virtual Tour URL, Floor Plan PDF/Image.
- **Lister/Agent Profile:** Nested object containing Agent Name, Avatar, Agency, Phone, Email, and License Number.

4. App Router Folder Structure

Organized for Next.js App Router, isolating route groups and enforcing modular component design.

```

my-real-estate-saas/
├── src/
│   ├── app/
│   │   ├── (public)/                # Public UI & Search Engine
│   │   │   ├── properties/
│   │   │   ├── agents/
│   │   │   └── auth/
│   │   ├── (dashboard)/            # Multi-Tenant Portals
│   │   │   ├── client/              # Buyer/Tenant Views
│   │   │   ├── agent/               # Landlord/Broker Views
│   │   │   └── (admin)/              # Global Admin Controls
│   │   │       ├── users/
│   │   │       ├── properties/
│   │   │       └── plans/
│   ├── components/
│   │   ├── ui/                     # Shadcn Primitives (Buttons, Toasts)
│   │   ├── maps/                   # Mapbox GL / Supercluster layers
│   │   ├── forms/                  # Multi-step property creation engine
│   │   └── layout/                  # Sidebars, Navbars, Footers
│   ├── lib/
│   │   ├── utils.ts                # Tailwind merge, formatting
│   │   └── formatters.ts           # Currency & dimension utilities
│   └── mocks/
│       ├── propertiesMock.ts        # High-fidelity mock property schema
│       └── usersMock.ts

```

This specification outlines the Frontend Prototype Phase (Weeks 1-3). Once verified, Backend integration (MongoDB, Meilisearch, Stripe) will commence.